

Data Analyst

. Analytics Foundations and Career Roadmap

- **YouTube:**

- [Alex The Analyst - Data Analyst Boot Camp](#) – The ultimate industry baseline mapping day-to-day responsibilities, expectations, and domain fundamentals.

- [Luke Barousse - How to Become a Data Analyst](#) – Exceptionally practical guides tracking current market tool stacks, project scoping, and portfolio strategies.

- **Hands-on Labs:**

- [Google Data Analytics Professional Certificate \(Coursera\)](#) – Foundational sandbox tracks introducing analytical thinking patterns and case studies.

- **Documentation:**

- [The Model Thinker Book / Guides](#) – Comprehensive overview detailing structural framework thinking, problem decomposition, and KPIs.

2. SQL for Data Retrieval and Transformation

- **YouTube:**

- [Luke Barousse - SQL Playlist for Analytics](#) – Focused explicitly on querying data for insights rather than database maintenance or engineering.

- [Alex The Analyst - Advanced SQL Tutorials](#) – Pristine walk-throughs of subqueries, window functions, CTEs, and table joins.

- **Hands-on Labs:**

- [SQLBolt](#) – Highly interactive, in-browser SQL training modules moving from basic queries to complex transformations.

- [DataLemur](#) – Practical SQL interview problems sourced directly from elite tech companies, targeted at analytical problem solving.

- **Documentation:**

- [PostgreSQL Official Docs \(Queries\)](#) & [BigQuery SQL Reference](#) – Prescriptive syntax handbooks for handling heavy aggregate metrics.

3. Python and Data Manipulation Libraries

- **YouTube:**

- [Keith Galli's Pandas Playlist](#) – High-yield, clear implementation guides sorting, filtering, merging, and cleaning messy data frames.

- [Corey Schafer - Pandas Playlist](#) – Deep, thorough dive into data slicing,

parsing datetime strings, and aggregate data transformations.

- **Hands-on Labs:**

- **Kaggle Learn: Pandas Track** – In-browser sandbox environments built around restructuring multidimensional tabular inputs.

- **DataCamp: Data Manipulation with Pandas** – Live scripting environments focused on matrix indexing and data cleaning functions.

- **Documentation:**

- **Pandas User Guide** – The definitive structural optimization handbook for operational analytical scripts.

4. Descriptive Statistics and Data Exploration (EDA)

- **YouTube:**

- **StatQuest with Josh Starmer - Statistics Fundamentals** – Unmatched visual breakdowns explaining mean, median, standard deviation, and data distributions simply.

- **Khan Academy - College Statistics Playlist** – Rigorous, direct conceptual walkthroughs covering variance, quartiles, and data shapes.

- **Hands-on Labs:**

- **Kaggle Datasets** – Proving grounds where you can pull raw, messy tabular matrices and script descriptive profiles from scratch.

- **Google Sheets / Excel Advanced Data Toolsets** – Build immediate confidence calculating covariance, distributions, and skew profiles dynamically.

- **Documentation:**

- **NIST Handbook of Statistical Methods** – The definitive baseline text detailing standard statistical explorations and baseline assumptions.

5. Data Visualization and Storytelling

- **YouTube:**

- **Storytelling with Data (Cole Nussbaumer Knaflic)** – Master-class deep dives explaining how to eliminate visual clutter and highlight key business metrics.

- **Chandoo - Data Visualization Playbooks** – Brilliant walkthroughs of clean chart formatting, visual design patterns, and context setting.

- **Hands-on Labs:**

- **Tableau Public Gallery & Challenges** – Free canvas to replicate and experiment with world-class, production-ready interactive visual maps.

- **MakeoverMonday Community Challenges** – Weekly data drops challenging you to redesign weak graphs into strong narrative pipelines.

- **Documentation:**

- **Seaborn Example Gallery & The Data Visualisation Catalogue** – Graphic reference dictionaries identifying the exact plot types matching specific data logic relationships.

6. Business Reporting and Dashboarding

- **YouTube:**

- **Avi Singh - PowerBI Tutorials** – Complete end-to-end framework covering data modeling, DAX parameters, and building clean report sheets.

- **SQLBI (Marco Russo & Alberto Ferrari)** – The ultimate global authority channel on optimizing deep financial dashboard calculations and calculations logic.

- **Hands-on Labs:**

- **Microsoft Power BI Guided Learning Tracks** – Free local environment testing paths focused on dimensional modeling (Star Schemas) and refresh loops.

- **Google Looker Studio Tutorials** – Cloud metrics labs matching dynamic live databases directly to automated business summaries.

- **Documentation:**

- **The Data Warehouse Toolkit (Ralph Kimball Book)** – The definitive global blueprint detailing facts, dimensions, and transactional data reporting structures.

7. Hypothesis Testing, Correlation, and Validation

- **YouTube:**

- **StatQuest - Hypothesis Testing & P-Values Explained** – Visual blocks mapping out T-tests, ANOVA, and Chi-Square evaluations cleanly.

- **Brandon Foltz - Statistics Playlist** – Thorough, methodical mathematical deep-dives into regression confidence boundaries and error types.

- **Hands-on Labs:**

- **OpenIntro Statistics Interactive Exercises** – Free problem sets designed to build immediate comfort with significance intervals and test boundaries.

- **SciPy Statistical Labs (Python Notebooks)** – Programmatic sandboxes applying real correlation and parameter tests to tabular metrics feeds.

- **Documentation:**

- **Statsmodels Documentation (Python)** – Detailed technical outlines covering regressions, mathematical diagnostics, and data variances.

8. Machine Learning Foundations for Analysts

- **YouTube:**

- **StatQuest - Linear Regression & Logistic Regression and LR** – Simple, unforgettable graphic explanations of algorithmic trends and predictive lines.

- **Machine Learning for Data Analysts** – Tailored explicitly for analysts, tracking

how to execute forecasting models without drowning in complex calculus.

- **Hands-on Labs:**

- [Google Developers ML Crash Course](#) – Fast interactive browser loops teaching optimization functions and metrics splits.

- [Scikit-Learn Regression Examples](#) – Simple copy-paste scripts modeling time-series forecasting or value estimations.

- **Documentation:**

- [Scikit-Learn User Guide \(Supervised Modules\)](#) – The undisputed programming manual detailing cluster logic and trend analysis.

9. Big Data Concepts and Data Platforms

- **YouTube:**

- [FreeCodeCamp - Data Engineering & Big Data Fundamentals](#) – Understand data lakes, cloud warehouses, storage clusters, and ETL logic.

- [Seattle Data Guy](#) – High-utility industry perspective outlining how massive datasets shift from storage blocks into analyst tables.

- **Hands-on Labs:**

- [AWS Workshop Studio \(Analytics / Cloud Practitioner\)](#) – Free structural templates mapping multi-source feeds into AWS Athena or Redshift.

- [Databricks Community Edition Sandbox](#) – Free cloud cluster access to practice executing heavy Spark queries across millions of text records.

- **Documentation:**

- [Snowflake Architecture Manual](#) & [Google BigQuery Architecture Guides](#) – Production playbooks detailing cloud data distribution, storage costs, and query sizing.

10. Portfolio Practice, Kaggle, and Continuous Analytics Delivery

- **YouTube:**

- [Alex The Analyst - Portfolio Project Series](#) – Step-by-step blueprints taking a raw dataset, cleaning it via SQL, and delivering a live, interactive dashboard.

- [Ken Jee - Data Science/Analytics Project Workflows](#) – Clear guides framing your projects logically to gain maximum attention on GitHub and LinkedIn.

- **Hands-on Labs:**

- [Kaggle Getting Started Competitions](#) – Proving ground to post notebooks, gather upvotes, and study solution scripts from senior developers.

- [GitHub Pages Setup Guides](#) – Build a fast, lightweight, and modern portfolio site completely hosted in your repository space.

- **Documentation:**

- **The Analytics Analytics Book Standards** – Corporate strategy frameworks tracking how to translate metrics insights into clear profit decisions.